Highlights

Rethinking Decoder Computation in 3D Medical Image Segmentation: A Characterization Study

Author One, Author Two, Author Three

- Characterizes the spatial contribution of additional decoder computation in 3D segmentation.
- Establishes the distinction between correctness activity (gross changes) and net gain (directional benefit).
- Demonstrates substantial bidirectional correctness changes with small net gain alongside extreme boundary concentration ($K_{80} \leq 5\%$).
- Demonstrates predictive uncertainty signals activity but provides weak directional discrimination.
- Evaluates retrospective output-space hybridization to recover macro Dice gaps without over-extrapolating runtime speedups.

Rethinking Decoder Computation in 3D Medical Image Segmentation: A Characterization Study

Author One[a,*], Author Two[b], Author Three[a]

[a]*Department Name, University Name, Address Line, City Name, 00000, State, Country*
[b]*Another Department, Another University, Address Line, City Name, 00000, State, Country*

Abstract

Modern 3D medical image segmentation networks (e.g. vision transformers and large-kernel CNNs) increasingly incorporate heavy multi-resolution decoders, assuming that dense decoding is universally necessary across the volumetric grid. Conversely, efficient alternatives (e.g. EffiDec3D and EfficientMedNeXt) simplify these decoding pathways based on macroscopic metrics. However, where, under what spatial conditions, and to what extent additional decoder computation actually contributes to segmentation correctness remain uncharacterized. We present a systematic characterization of decoder computation in 3D medical image segmentation. We evaluate end-to-end full–efficient decoder pairs across three representative backbones on BTCV CT, examine cross-dataset consistency using 3D UX-Net across BTCV CT, MSD01 BrainTumour MRI, and MSD08 HepaticVessel CT, and perform controlled factor attribution using a frozen shared-encoder 2×2 factorial design on 3D UX-Net. Our findings establish a core conceptual distinction: *correctness activity* measures where the decoder alters prediction correctness (gross activity), whereas *decoder net gain* measures the directional balance between positive voxel corrections and negative degradations. Across architectures and datasets, additional decoder capacity produced substantial bidirectional correctness activity, whereas the resulting net shift was much smaller and task-dependent. Positive gain was strongly spatially concentrated, with an oracle selector capturing at least 80% of positive gain within the top 5% of spatial blocks ($K_{80} \leq 5\%$). Predictive entropy localized correctness-changing regions substantially better than it discriminated whether those changes were beneficial, yielding Location AUROCs of 0.865–0.913 and Direction AUROCs of 0.520–0.591. Retrospective output substitution within a 20% entropy-selected block budget recovered the full macro-Dice gap for 3D UX-Net and SwinUNETR, while recovery remained partial for MedNeXt-M-K3. These results show that prediction difficulty and computational utility are related but distinct quantities, offering empirical bounds and design directions for future adaptive decoders.

Keywords: 3D Medical Image Segmentation, Efficient Deep Learning, Uncertainty Quantification, Decoder Computation, Model Characterization

1. Introduction

Precise 3D medical image segmentation is essential for clinical applications such as image-guided surgical planning, volumetric tumor monitoring, and targeted radiation therapy. In deep learning, encoder-decoder networks based on the U-Net paradigm [1, 2] remain the foundational standard. Recent research has primarily focused on scaling encoder expressiveness through vision transformers, large convolutional kernels, and self-supervised pre-training [3, 4, 5, 6]. In parallel, modern 3D architectures retain heavy multi-resolution decoders equipped with dense upsampling stages, complex skip-fusion connections, and spatial refinement blocks. Because 3D volumetric scans scale cubically in resolution ($O(N^3)$), these multi-scale decoders routinely account for over 40% of total inference FLOPs, posing severe memory and latency bottlenecks for real-time intraoperative deployment. This prevailing architectural trajectory implicitly assumes that dense, high-capacity decoding is universally necessary across every voxel in the 3D volume.

Conversely, emerging lightweight models attempt to mitigate this computational burden by simplifying decoding pathways based on macroscopic metrics, such as EffiDec3D [7] and EfficientMedNeXt [8]. However, standard evaluation paradigms assess decoder designs almost exclusively through aggregate metrics such as the global Dice similarity coefficient or Hausdorff distance. While these macroscopic metrics summarize overall performance, they average out localized prediction changes and conceal spatial micro-dynamics across homogeneous organ interiors vs. complex anatomical boundaries. Consequently, a fundamental scientific question remains unexamined: **How does additional decoder computation actually contribute to 3D medical image segmentation?**

To address this question, we systematically characterize decoder computation in 3D medical image segmentation models. Rather than proposing a new network architecture, we treat computation characterization as a distinct empirical approach. The main contributions of this study are:

1. We introduce a voxel-level framework that separates de-

*Corresponding author
Email address: `author1@institute.edu` (Author One)

coder correctness activity from directional net gain.

2. We characterize decoder contribution in terms of spatial heterogeneity, positive-gain concentration, and inference-time predictability.
3. We provide controlled evidence across architectures, datasets, and decoder interventions, and establish retrospective opportunity bounds for future utility-aware adaptive decoding.

The remainder of this paper is organized as follows: Section 2 reviews related literature and highlights key gaps. Section 3 details our characterization framework and metrics. Section 4 describes our experimental setup and evaluation protocols. Section 5 presents our observational findings, Section 6 discusses underlying mechanisms and limitations, and Section 7 concludes the paper.

2. Related Work

2.1. Decoder Design in 3D Medical Image Segmentation

The encoder-decoder architecture introduced by U-Net [1, 2] and V-Net [9] remains the central design pattern for 3D medical image segmentation. Frameworks such as nnU-Net [10] demonstrated the effectiveness of standardized training protocols combined with deep decoding pathways. Subsequent developments incorporated Transformer backbones and large-kernel convolutions to capture long-range contextual dependencies (e.g., TransUNet [11], UNETR [3], Swin UNETR [4], 3D UX-Net [5], and MedNeXt [6]). These modern architectures emphasize expressive feature extraction while maintaining dense multi-scale decoders. However, these designs are evaluated primarily through final segmentation accuracy and overall parameter count; how decoder capacity operates spatially across volume regions remains unexamined.

2.2. Efficient Segmentation Architecture Design

To mitigate the high computational cost of 3D volumetric convolutions, recent works have explored lightweight decoder designs. EffiDec3D [7] introduced systematic decoder pruning by reducing channel dimensions and eliminating high-resolution decoding stages, establishing trade-offs between parameter scale and Dice performance. Similarly, EfficientMedNeXt [8] simplified decoding stages before introducing residual blocks. Other lightweight models achieve efficiency through hybrid transformer-convolution encoders, attention pruning, or custom lightweight modules (e.g., SegFormer [12], UNETR++ [13], Slim UNETR [14]). Nevertheless, efficient architecture studies infer redundancy indirectly from global performance curves, leaving unexamined whether the eliminated computation was spatially uniform or concentrated in specific anatomical sub-regions.

2.3. Model Characterization and Uncertainty Quantification

Recent studies have explored adaptive inference for volumetric processing [15, 16]. Epistemic and aleatoric uncertainty metrics have been widely applied to quantify model confidence and detect out-of-distribution inputs [17, 18, 19]. Existing uncertainty-aware quality assessment methods estimate whether a current segmentation is likely to be accurate. In contrast, our study asks a counterfactual computational question: whether allocating additional decoder computation is likely to improve or degrade that segmentation. Uncertainty-driven dynamic execution in 3D medical segmentation is hampered by a critical unexamined gap: it remains unclear whether predictive uncertainty can locate regions where additional decoder computation alters correctness, and whether it can further distinguish beneficial corrections from harmful degradations. Our framework explicitly addresses these questions by separating correctness activity from directional net gain.

3. Methods: A Characterization Framework

To understand the spatial and functional contribution of decoder computation, we establish a controlled observation framework (Figure 1). This section outlines the controlled decoder interventions, voxel-level correctness framework, rate definitions, spatial block partitioning, and characterization metrics.

3.1. Controlled Decoder Interventions

We analyze decoder capacity by isolating two generic architectural factors that govern volumetric decoding computation:

1. **Channel Capacity ($C_{\mathbf{dec}}$)**: The feature channel dimension across upsampling layers, controlling feature representation width ($C_{\text{full}} \to C_{\text{reduced}}$).
2. **High-Resolution Decoder Stages (RF)**: The resolution scaling factor, controlling whether high-resolution upsampling stages (e.g., full resolution 1×) are retained (RF_1) or omitted (RF_2).

Crossing these two experimental factors forms a 2×2 factorial intervention design (Full E_0, Channel-reduced, Stage-reduced, and Combined-efficient E_1). The specific channel-reduction ratios and stage-removal configurations follow the architecture-preserving decoder simplification strategy introduced by EffiDec3D [7]. Interventions occur strictly within the decoder pathway (upsampling, skip fusion, and refinement) while keeping the feature encoder architecture untouched.

Choice of Intervention Template: We adopt the reduction principles of EffiDec3D as a controlled intervention template rather than treating EffiDec3D itself as the primary object of evaluation. Its decoder-localized and architecture-preserving modifications provide matched configurations in which channel capacity and high-resolution stage retention can be independently controlled. This enables spatial characterization of decoder contribution while minimizing confounding changes to the encoder family, prediction task, or training objective.

3.2. Voxel-Level Correctness Activity Framework

Let v denote a voxel within a 3D medical image volume V, with ground truth label $y(v)$, full decoder prediction $\hat{y}_f(v)$, and efficient decoder prediction $\hat{y}_e(v)$. Comparing predictions between matched E_0 and E_1 decoders yields two correctness-changing outcomes per voxel (Figure 2; see Supplementary

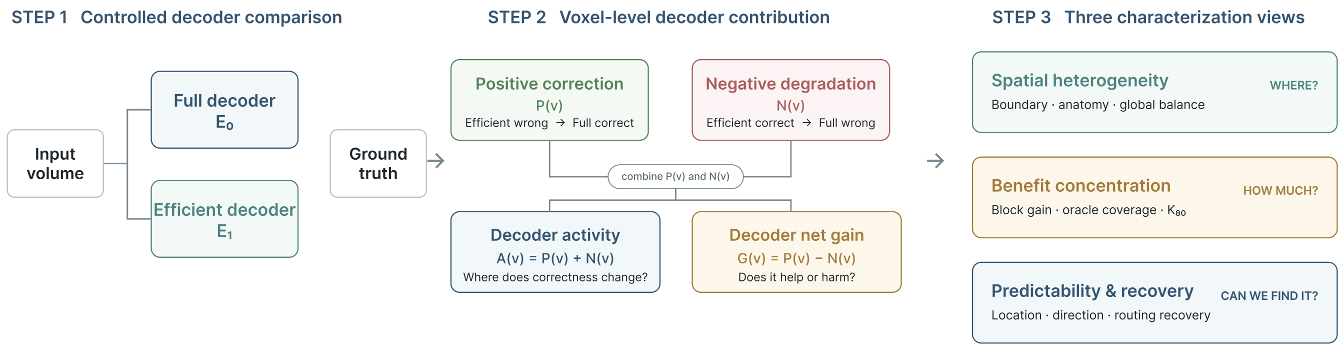


Figure 1: Overview of the decoder computation characterization framework across Step 1 (Controlled Interventions), Step 2 (Voxel Correctness Activity $A_{\text{corr}}(v)$ and Net Gain $G(v)$), and Step 3 (Three Characterization Axes).

Figures S1 and S2 for 3D UX-Net and MedNeXt-M-K3 visualizations):

- **Positive Correction** ($P(v)$): $P(v) = \mathbf{1}[\hat{y}_e(v) \neq y(v) \wedge \hat{y}_f(v) = y(v)]$, where the full decoder corrects an efficient baseline error.
- **Negative Degradation** ($N(v)$): $N(v) = \mathbf{1}[\hat{y}_e(v) = y(v) \wedge \hat{y}_f(v) \neq y(v)]$, where the full decoder alters a previously correct efficient prediction.

Defining voxel correctness indicators $C_f(v) = \mathbf{1}[\hat{y}_f(v) = y(v)]$ and $C_e(v) = \mathbf{1}[\hat{y}_e(v) = y(v)]$, we formalize two fundamental voxel indicators:

- **Decoder Net Gain** ($G(v) = C_f(v) - C_e(v) = P(v) - N(v)$): Measures the net directional balance ($G(v) = +1$ for correction, -1 for degradation, 0 for unchanged).
- **Correctness Activity** ($A_{\mathbf{corr}}(v) = P(v) + N(v) = |G(v)|$): Measures where additional decoder capacity alters prediction correctness (gross activity).

3.3. Subject-Level Rate Formulations and Evaluated Domains

To aggregate voxel-level indicators into global rates for subject i, we define the subject-level positive correction rate $R_{\text{pos}}^{(i)}$, negative degradation rate $R_{\text{neg}}^{(i)}$, gross correctness activity rate $R_{\text{act}}^{(i)}$, and net gain rate $R_{\text{net}}^{(i)}$ over an evaluation domain Ω_i:

$$R_{\text{pos}}^{(i)} = \frac{1}{|\Omega_i|} \sum_{v \in \Omega_i} P_i(v), \quad R_{\text{neg}}^{(i)} = \frac{1}{|\Omega_i|} \sum_{v \in \Omega_i} N_i(v) \tag{1}$$

$$R_{\text{act}}^{(i)} = R_{\text{pos}}^{(i)} + R_{\text{neg}}^{(i)}, \quad R_{\text{net}}^{(i)} = R_{\text{pos}}^{(i)} - R_{\text{neg}}^{(i)} \tag{2}$$

We evaluate rates across two explicit spatial domains Ω_i:

1. **All-Volume Domain** ($\Omega_{\mathbf{all}}$): All voxels within the 3D bounding volume grid V_i, denoted by $R_{\text{act}}^{\text{all}}$ and $R_{\text{net}}^{\text{all}}$.
2. **Foreground-Union Domain** ($\Omega_{\mathbf{fg}}$): The subset of voxels belonging to ground-truth or predicted foreground structures: $\Omega_{\text{fg}} = \{v \in V_i \mid y_i(v) > 0 \vee \hat{y}_{e,i}(v) > 0 \vee \hat{y}_{f,i}(v) > 0\}$, denoted by $R_{\text{act}}^{\text{fg}}$ and $R_{\text{net}}^{\text{fg}}$.

Global metrics are reported as subject-level population means with 95% confidence intervals computed via 1000-resample paired bootstrapping across subjects.

3.4. Characterization Metrics and Spatial Partitioning

We evaluate decoder contribution across three core analytical axes (Table 1):

3.4.1. Physical Ground-Truth Anatomical Boundary Distance

To prevent circularity from model prediction boundaries, spatial distance analysis is anchored strictly to the **Ground-Truth Anatomical Boundary** using physical Euclidean distances. Incorporating dataset voxel spacings $\mathbf{D} = \text{diag}(s_x, s_y, s_z)$, physical distance $d_{GT}(v)$ is defined by:

$$d_{GT}(v) = \min_{u \in \partial S_{GT}} ||\mathbf{D}(x_v - x_u)||_2 \tag{3}$$

where ∂S_{GT} represents the boundary voxel set of ground-truth annotations $\bigcup_c (y(v) = c)$. Correctness activity $A_{\text{corr}}(v)$ and net gain $G(v)$ are stratified across physical distance bands (0–2 mm, 2–4 mm, 4–8 mm, > 8 mm).

3.4.2. Spatial Block Construction and Physical Scale

For block-level spatial analysis, volumes are resampled to dataset-specific target spacings (BTCV: $1.5 \times 1.5 \times 2.0$ mm^3; MSD01: $1.0 \times 1.0 \times 1.0$ mm^3; MSD08: $0.8 \times 0.8 \times 1.5$ mm^3). To ensure cross-dataset comparability, block dimensions are constructed to match physical dimensions of approximately $24 \times 24 \times 24$ mm (BTCV: $16 \times 16 \times 12$ voxels; MSD01: $24 \times 24 \times 24$ voxels; MSD08: $30 \times 30 \times 16$ voxels). Block activity $A_{\text{corr}}(r) = \mathbf{1}[\sum_{v \in b_r} A_{\text{corr}}(v) > 0]$ detects active blocks. For directional utility classification, blocks with non-zero net gain $G(r) = \sum_{v \in b_r} G(v) \neq 0$ are evaluated for direction AUROC, excluding zero-gain blocks. Block-level metrics (K_{80}, Location AUROC, Direction AUROC) were computed within each subject and summarized across subjects, with confidence intervals estimated using clustered bootstrap resampling at the subject level.

3.4.3. Positive Gain Concentration (K_{80})

Sorting blocks by $G(r)$, oracle coverage is defined by:

$$Coverage(k) = \frac{\sum_{r=1}^{\lfloor (k/100) \cdot |B| \rfloor} \max(G(r), 0)}{\sum_{r=1}^{|B|} \max(G(r), 0)} \tag{4}$$

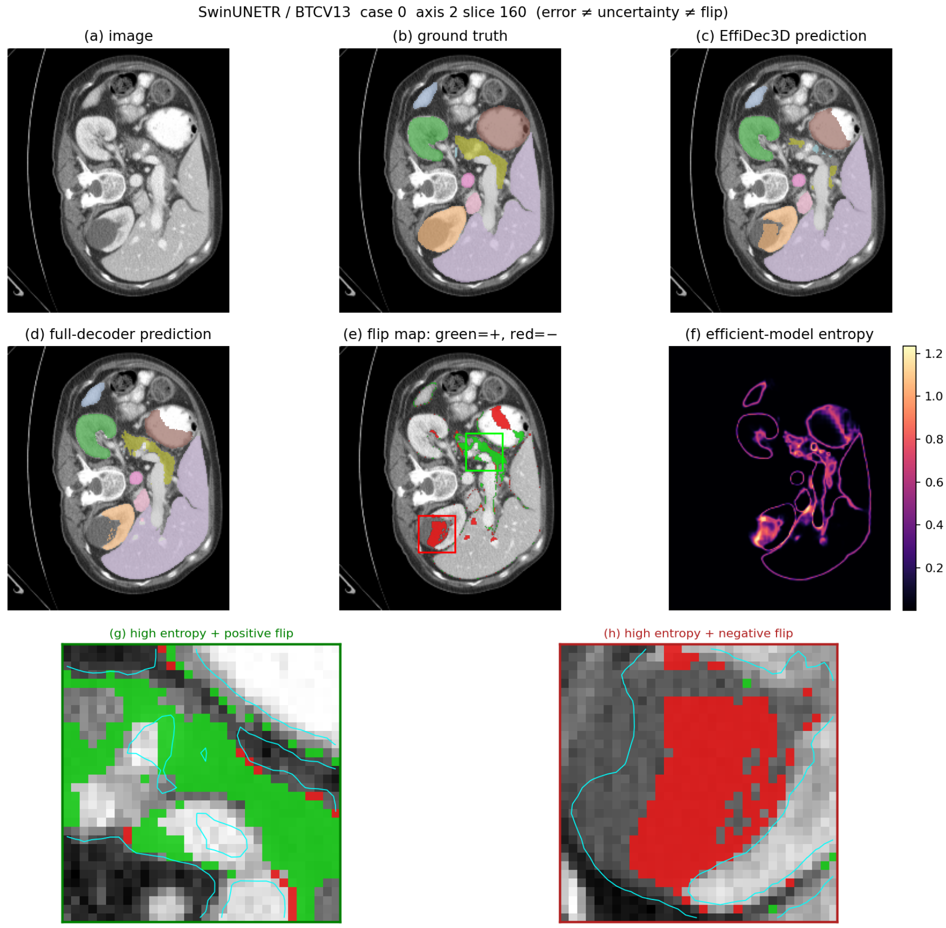


Figure 2: Qualitative illustration of correctness activity and directional gain for SwinUNETR on BTCV CT dataset under Full (E_0) vs. Efficient (E_1) decoders (see Supplementary Figures S1 and S2 for 3D UX-Net and MedNeXt-M-K3).

Table 1: Overview of characterization objectives and evaluation measures.

Analysis Objective	**Evaluation Measure**	**Question Answered**
Gross spatial effect	Correctness Activity $A_{\mathrm{corr}}(v)$	Where does additional decoder computation change prediction correctness?
Directional effect	Net Gain $G(v)$	Where does the decoder improve correctness overall?
Benefit concentration	Oracle K_{80}	How spatially concentrated is positive net gain?
Activity predictability	Location AUROC	Can cheap signals detect where correctness changes occur?
Gain predictability	Direction AUROC	Can cheap signals distinguish beneficial from harmful changes?
Retrospective recovery	Retrospective Hybrid $R_{\mathrm{recovery}}(k)$	How much of the full decoder's Dice gap is recovered by output hybridization?
Multi-sample uncertainty	TTA Mutual Info $I(v)$	Does multi-view epistemic disagreement resolve direction predictability?

We record K_{80}, defined as the minimum percentage of spatial blocks required by an oracle selector to contain 80% of the total positive decoder gain [20].

3.4.4. *Retrospective Output-Space Hybridization ($R_{recovery}$)*

To evaluate whether boundary-localized decoder improvements account for macro-metric gains, we evaluate a retrospective output-space hybrid prediction $\hat{y}_k(v)$, which substitutes full decoder predictions $\hat{y}_f(v)$ into selected blocks S_k while retaining efficient baseline predictions $\hat{y}_e(v)$ elsewhere:

$$\hat{y}_k(v) = \begin{cases} \hat{y}_f(v), & \text{if } v \in S_k \\ \hat{y}_e(v), & \text{otherwise} \end{cases} \tag{5}$$

When $\mathrm{Dice}(\hat{y}_f) - \mathrm{Dice}(\hat{y}_e) > 0$, we measure retrospective Dice gap recovery fraction $R_{\mathrm{recovery}}(k)$:

$$R_{\mathrm{recovery}}(k) = \frac{\mathrm{Dice}(\hat{y}_k) - \mathrm{Dice}(\hat{y}_e)}{\mathrm{Dice}(\hat{y}_f) - \mathrm{Dice}(\hat{y}_e)} \tag{6}$$

If the full decoder gap is non-positive ($\mathrm{Dice}(\hat{y}_f) \leq \mathrm{Dice}(\hat{y}_e)$), absolute Dice difference $\Delta\mathrm{Dice}(k) = \mathrm{Dice}(\hat{y}_k) - \mathrm{Dice}(\hat{y}_e)$ is reported instead. Values $R_{\text{recovery}}(k) > 1.0$ indicate over-recovery and are retained unclipped.

3.4.5. Uncertainty Signals and Predictability Scores

We compute predictive Shannon entropy $H(v) = -\sum_{c=1}^{C} p_c(v) \log p_c(v)$ from efficient baseline probabilities [21, 17, 22]. Block-level predictor scores $H(r)$ and Test-Time Augmentation Mutual Information (TTA-MI) $I(r)$ are defined by block volume means:

$$H(r) = \frac{1}{|b_r|} \sum_{v \in b_r} H(v), \quad I(r) = \frac{1}{|b_r|} \sum_{v \in b_r} I(v) \tag{7}$$

where $I(v) = H(\bar{p}(v)) - \frac{1}{K} \sum_{k=1}^{K} H(p_k(v))$ over $K = 8$ spatial view transformations. All TTA probability maps were inverse-warped to the original image coordinate system before computing ensemble means and mutual information. Predictability is benchmarked across: (i) **Location AUROC/AUPRC** ($H(r)$ predicting $A_{\text{corr}}(r) > 0$), (ii) **Direction AUROC/AUPRC** ($H(r)$ predicting $G(r) > 0$ on non-zero gain blocks) [23], and (iii) **Conditional Density Overlap** $p(H \mid P(v) = 1)$ vs. $p(H \mid N(v) = 1)$.

3.4.6. Factorial Main Effects and Interaction Analysis

In our 2×2 decoder configuration grid, we quantify the channel main effect Δ_C, stage main effect Δ_{RF}, and channel-stage interaction $\Delta_{C \times RF}$ for any outcome metric M:

$$\begin{aligned} \Delta_C = \frac{1}{2}(&M(C_{48}, RF_1) + M(C_{48}, RF_2) \\ &- M(C_{384}, RF_1) - M(C_{384}, RF_2)) \end{aligned} \tag{8}$$

$$\begin{aligned} \Delta_{RF} = \frac{1}{2}(&M(C_{384}, RF_2) + M(C_{48}, RF_2) \\ &- M(C_{384}, RF_1) - M(C_{48}, RF_1)) \end{aligned} \tag{9}$$

$$\begin{aligned} \Delta_{C \times RF} =& M(C_{48}, RF_2) - M(C_{48}, RF_1) \\ &- M(C_{384}, RF_2) + M(C_{384}, RF_1) \end{aligned} \tag{10}$$

4. Experimental Setup

This section describes the target 3D backbones, clinical datasets, evaluation axes, and implementation setup.

4.1. Target Architectures and Intervention Specifications

We evaluate matched decoder pairs across three representative 3D medical image segmentation backbones: a large-kernel CNN (3D UX-Net [5]), a vision transformer (SwinUNETR [4]), and a ConvNeXt-style network (MedNeXt-M-K3 [6]). Interventions occur strictly within the decoder pathway (upsampling, skip fusion, and refinement) while keeping feature encoder architectures untouched. Table 2 details the intervention specifications across backbones.

Architectural & FLOP Profiling Protocol: All computational complexity metrics (FLOPs/MACs) were measured using MONAI's 'fvcore' profiling utility on a single 3D patch input tensor of shape $1 \times C \times 96 \times 96 \times 96$. To prevent cross-table confusion, we explicitly distinguish two specific 3D UX-Net full configurations evaluated in this study:

1. **End-to-End Reference Configuration** (Table 3): Trained end-to-end with default MONAI training pipelines containing auxiliary deep-supervision heads (53.01M params / 578.74 GMac).
2. **Frozen-Encoder Factorial Configuration** (Table 4): Evaluates the standalone decoder pathway with auxiliary heads detached under fixed unpadded patch profiling ($96 \times 96 \times 96$, 46.72M params / 657.76 GMac).

4.2. Factorial Configurations

To cleanly isolate the architectural mechanisms driving decoder activity, we perform a 2×2 factorial analysis on the 3D UX-Net backbone. By freezing the encoder and separating the decoding capacity along two structural axes (feature resolution and channel width), we evaluate whether performance and error variations originate from high-resolution spatial routing or abstract semantic capacity. Note that the factorial full configuration uses a separately instantiated decoder pathway to isolate these specific variables, and its FLOPs (657.76 GMac) are therefore not directly comparable with the tightly integrated end-to-end full configuration (578.74 GMac).
End-to-End Reference Full (Table 2): The baseline symmetric architecture.
Factorial-Analysis Full (Table 4): The isolated decoder pathway with all factorial features intact.

4.3. Clinical Datasets and Target Spacings

Experiments are conducted across three diverse 3D volumetric medical imaging datasets covering different anatomical regions and imaging modalities. Volumes are resampled to target spacings to match physical block side lengths of approximately 24 mm:

1. **BTCV Multi-Organ CT Dataset** [24]: 30 abdominal CT scans (resampled to $1.5 \times 1.5 \times 2.0$ mm^3; block dimensions $16 \times 16 \times 12$ voxels) with manual annotations for 13 organ structures.
2. **MSD01 BrainTumour MRI Dataset** [25, 26]: 484 multi-modal 3D MRI volumes (resampled to $1.0 \times 1.0 \times 1.0$ mm^3; block dimensions $24 \times 24 \times 24$ voxels) with 4 target classes.
3. **MSD08 HepaticVessel CT Dataset** [26]: 443 abdominal CT volumes (resampled to $0.8 \times 0.8 \times 1.5$ mm^3; block dimensions $30 \times 30 \times 16$ voxels) focused on vascular and neoplastic structures.

4.4. Experimental Evaluation Axes

We evaluate decoder contribution along three complementary experimental axes rather than a full architecture-by-dataset Cartesian grid:

1. **Cross-Architecture Consistency Axis**: Evaluates matched full (E_0) vs. efficient (E_1) decoders across three backbones (3D UX-Net, SwinUNETR, MedNeXt-M-K3) on the canonical BTCV multi-organ CT dataset to test architectural consistency.

Table 2: Backbone decoder specifications and efficient intervention parameters.

Architecture	Base Feature Size	Full Decoder (E_0)	Efficient Intervention (E_1)
3D UX-Net [5]	$C = 48$	C_{384}, 1× resolution	384 → 48 (87.5% cut), omit Stage 1
SwinUNETR [4]	$C = 48$	C_{384}, 1× resolution	384 → 48 (87.5% cut), omit Stage 1
MedNeXt-M-K3 [6]	$C = 32$	C_{256}, 1× resolution	256 → 32 (87.5% cut), omit Stage 1

2. **Cross-Dataset Consistency Axis**: Evaluates matched full (E_0) vs. efficient (E_1) decoders using a fixed canonical backbone (3D UX-Net) across three diverse datasets (BTCV CT, MSD01 BrainTumour MRI, and MSD08 HepaticVessel CT) to test modality and anatomical generalization.
3. **Decoder-Isolated Factor Attribution Axis**: Evaluates a 2×2 factorial intervention design (E_0, Channel-only, Stage-only, E_1) using a frozen shared 3D UX-Net encoder on BTCV CT. Holding encoder feature representations strictly constant ensures that prediction shifts are primarily attributable to specific decoder interventions (channel width vs. resolution scaling).

4.5. Training Protocols and Implementation Setup

All models were implemented in PyTorch using an NVIDIA RTX 5090 GPU with CUDA 12.8. Models were trained for 45,000 optimizer iterations using bfloat16 mixed precision and the AdamW optimizer (initial learning rate 1×10^{-3} with cosine annealing, weight decay 1×10^{-5}) under a compound Dice-CrossEntropy loss function, evaluating every 250 iterations to match the EffiDec3D protocol. Sub-volume patches of $96 \times 96 \times 96$ voxels were extracted with an optimizer batch size of 1 volume per GPU, drawing 4 spatial crops per volume (yielding an effective batch size of 4 crops per optimization step). Standard 3D online data augmentations (random rotations, flips, intensity scaling) and dataset-specific intensity normalization were applied. Evaluation networks dynamically allocated a foreground sliding-window overlap of 0.7.

4.6. Data Splits and Model Selection

Full and efficient configurations used identical subject-level training, validation, and test partitions. Model checkpoints were selected exclusively according to validation macro-Dice, and all subsequent characterization analyses were performed on held-out test subjects. Confidence intervals were estimated by subject-level bootstrap resampling over the held-out test set.

5. Results

In this section, we present observational characterization findings across our three experimental evaluation axes.

5.1. Accuracy–Computation Trade-off on BTCV

Table 3 summarizes aggregate computational complexity and segmentation performance under standard end-to-end model training across three 3D backbones on BTCV CT. For each backbone, we evaluate the standard Full decoder (E_0) against an Efficient baseline (E_1) implementing the parameter reduction strategy of EffiDec3D [7]. Across all three backbones, E_1 achieves substantial FLOP reductions (up to 91.4% in 3D UX-Net) while retaining competitive Mean Dice scores (e.g., MedNeXt-M-K3 E_0 achieves 82.61% Mean Dice vs. 81.35% for E_1, while E_1 reduces inference latency from 27.8 ms to 12.9 ms).

5.2. Correctness Activity and Net Gain across Architectures

Evaluating voxel-level prediction shifts between matched E_0 and E_1 decoders reveals substantial gross correctness activity $R_{\text{act}} = R_{\text{pos}} + R_{\text{neg}}$, whereas the net directional gain $R_{\text{net}} = R_{\text{pos}} - R_{\text{neg}}$ remains substantially smaller due to bidirectional cancellation (Figure 3).

Subject-mean All-Volume net gain rates $R^{\text{all}}_{\text{net}}$ across models are: +0.046% (95% CI: [−0.020%, +0.110%]) for 3D UX-Net, −0.001% (95% CI: [−0.045%, +0.040%]) for SwinUNETR, and +0.052% (95% CI: [−0.010%, +0.108%]) for MedNeXt-M-K3. Evaluated over the Foreground-Union domain Ω_{fg}, net gain rates are $R^{\text{fg}}_{\text{net}} = +0.381\%$ for 3D UX-Net, −0.008% for SwinUNETR, and +0.446% for MedNeXt-M-K3. Across the three evaluated model pairs, subject-bootstrap confidence intervals for global net gain included zero, indicating no statistically detectable directional shift. In all models, positive voxel corrections $P(v)$ are largely offset by negative degradations $N(v)$.

5.3. Spatial Heterogeneity

5.3.1. Physical Ground-Truth Boundary Concentration

Stratifying correctness activity $A_{\text{corr}}(v)$ by physical Euclidean distance $d_{GT}(v)$ to true ground-truth anatomical boundaries shows that correctness changes peak sharply within a 0–2 mm physical distance band (Figure 4). Both positive corrections and negative degradations decrease rapidly in organ interiors ($d_{GT} > 8$ mm).

5.3.2. Anatomical Structure Volatility

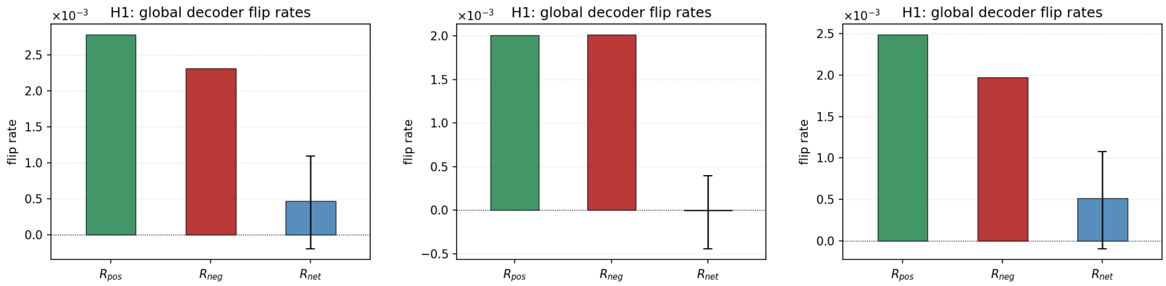
Anatomical analysis (Figure 5) demonstrates that small, morphologically complex organs exhibit high flip volatility. The left and right adrenal glands show positive correction rates of 16.2%/16.9% in 3D UX-Net, 14.4%/14.8% in SwinUNETR, and 12.3%/14.0% in MedNeXt-M-K3. Conversely, large structures like the liver remain highly stable (< 3% correctness activity).

5.4. Frozen-Encoder Factor Attribution

Table 4 reports our 2×2 factorial configuration grid on 3D UX-Net with frozen shared encoders on BTCV CT. Table 5 presents the main effects and interaction analysis across key outcome metrics:

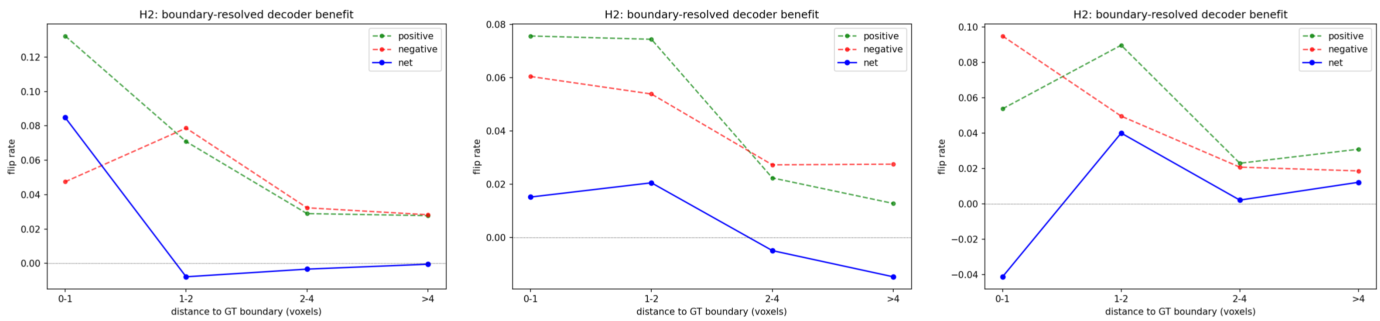
Table 3: Quantitative comparison of Full (E_0) vs. Efficient (E_1) decoders on the BTCV 13-organ dataset across three backbones (Cross-Architecture Consistency Axis).

Architecture	**Variant**	**Params (M)** ↓	**FLOPs (GMac)** ↓	**Latency (ms)** ↓	**Mean Dice (%)** ↑	**Mean HD95 (mm)** ↓
3D UX-Net	Full (E_0)	53.01	578.74	40.6	79.18	9.04
	Effi (E_1)	3.16	49.49	8.0	77.73	11.82
SwinUNETR	Full (E_0)	62.19	308.24	37.0	80.48	8.88
	Effi (E_1)	11.21	55.84	16.7	79.49	8.90
MedNeXt-M-K3	Full (E_0)	17.56	106.34	27.8	82.61	7.25
	Efficient (E_1)	5.80	49.54	12.9	81.35	6.77



(a) 3D UX-Net Global Flips (b) SwinUNETR Global Flips (c) MedNeXt-M-K3 Global Flips

Figure 3: Voxel-level net gain rates across 3D UX-Net, SwinUNETR, and MedNeXt-M-K3 backbones under Full (E_0) vs. Efficient (E_1) decoders.



(a) 3D UX-Net Boundary Flips (b) SwinUNETR Boundary Flips (c) MedNeXt-M-K3 Boundary Flips

Figure 4: Spatial ground-truth boundary concentration of prediction flips across the three evaluated backbones.

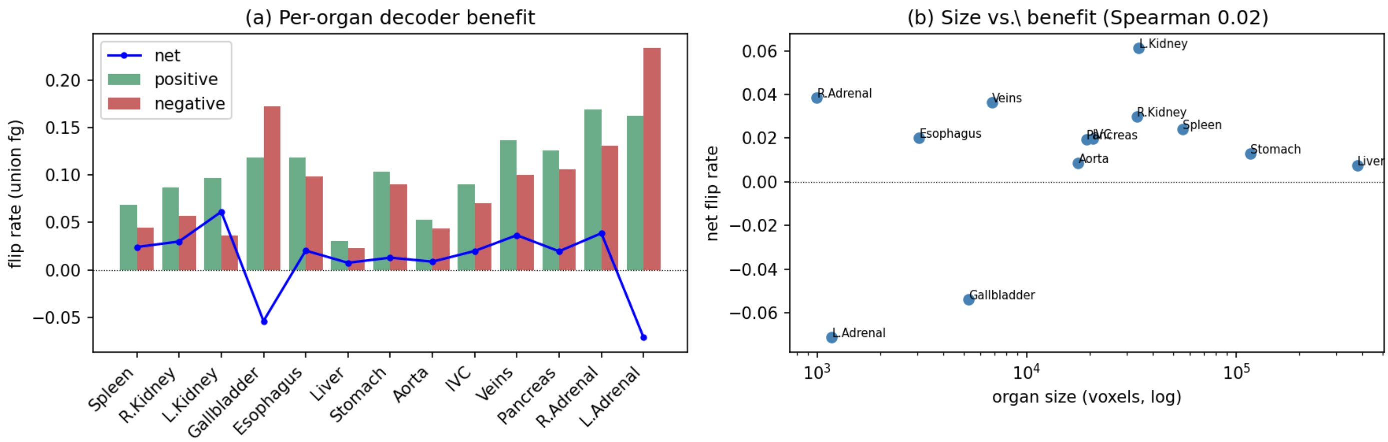
- **High-Resolution Stage Removal** ($RF_1 \to RF_2$): Removing the high-resolution stage accounts for 657.8 GMac → 319.1 GMac FLOP savings. Retaining this high-resolution stage yields a net positive correction rate of +0.033% in the 0–2 mm physical GT boundary band and +0.031% in the 2–4 mm band, decaying to negative net values (−0.011%) in organ interiors (> 8 mm).
- **Channel Width Reduction** ($C_{384} \to C_{48}$): Reducing decoder channels compresses parameter count by 95% (46.7M → 2.2M) while exerting a spatially diffuse and net-neutral effect ($R^{\text{all}}_{\text{net}} = +0.027\%$, 95% CI: $[-0.027\%, +0.083\%]$).

5.5. *Gain Concentration and Predictability*

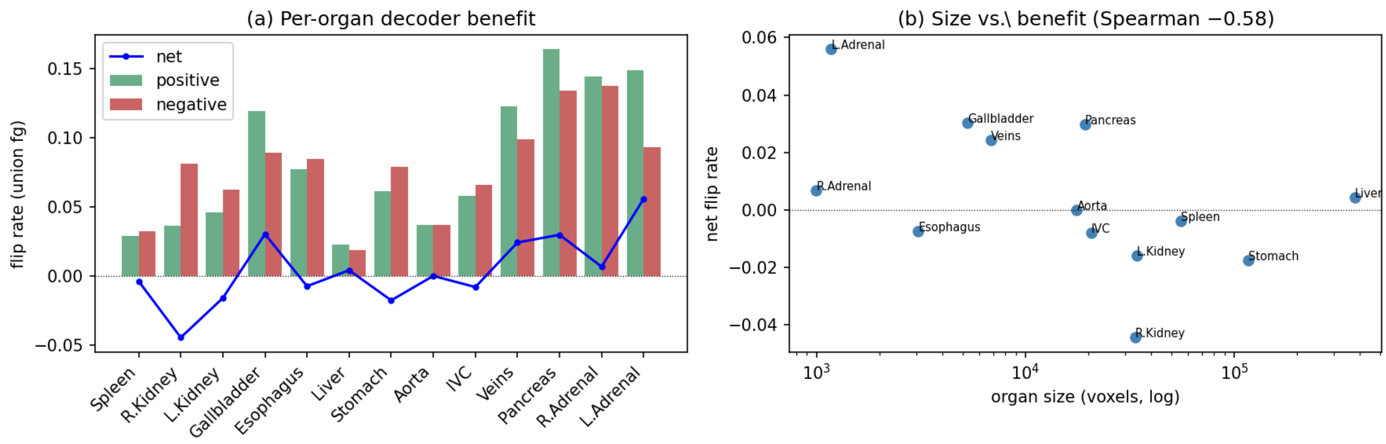
Figure 6 presents oracle benefit concentration curves. Across the three BTCV backbone pairs, the top 5% of spatial blocks captured at least 80% of positive gain ($K_{80} \leq 5\%$), and the top 10% captured up to 99.98%.

We evaluate whether cheap inference-time signals can locate regions where decoder computation alters correctness (Location AUROC) and whether they can further discriminate the direction of those changes (Direction AUROC) (Table 6 and Figure 7):

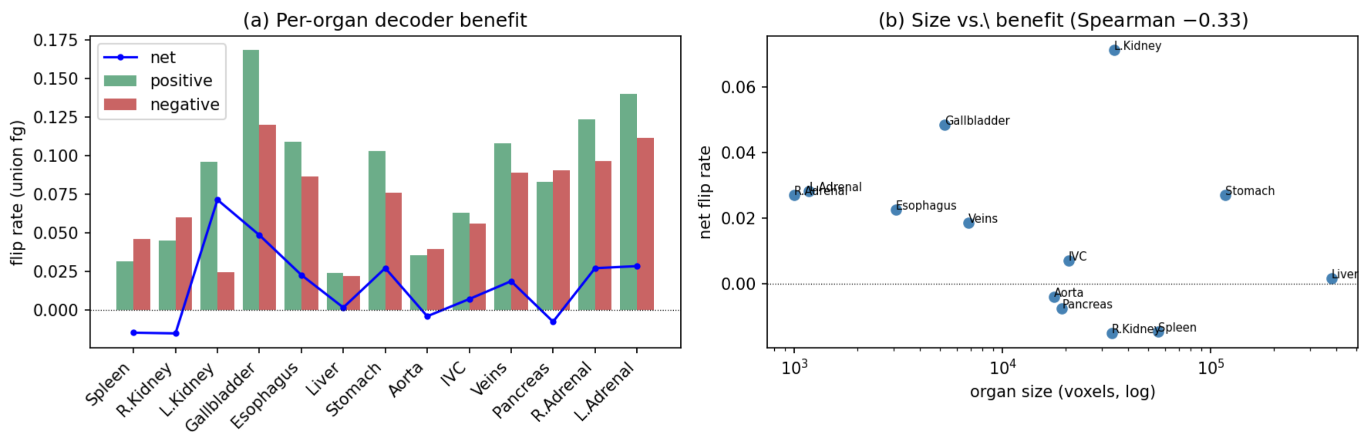
- **Activity Location AUROC**: Predictive entropy reliably identifies active blocks ($A_{\text{corr}}(r) > 0$), achieving high location AUROCs of 0.900 for 3D UX-Net (AUPRC 0.371), 0.913 for SwinUNETR (AUPRC 0.351), and 0.907 for MedNeXt-M-K3 (AUPRC 0.389).
- **Directional Utility Discrimination**: Predictive entropy provides strong activity localization but only weak directional discrimination ($G(r) > 0$ vs. $G(r) < 0$), yielding direction AUROCs modestly above chance: 0.591 for 3D



(a) 3D UX-Net Organ-Level Flips



(b) SwinUNETR Organ-Level Flips



(c) MedNeXt-M-K3 Organ-Level Flips

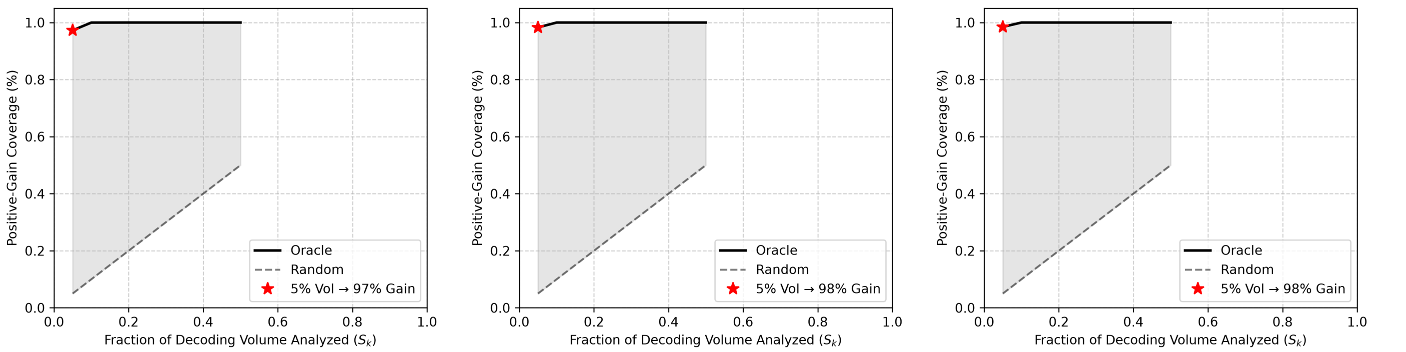
Figure 5: Organ-level flip breakdown across backbones on the BTCV CT multi-organ dataset under Full (E_0) vs. Efficient (E_1) decoders.

UX-Net, 0.560 for SwinUNETR, and 0.556 for MedNeXt-M-K3.

- **TTA Mutual Information (TTA-MI)**: Multi-sample epistemic disagreement $I(v)$ over $K = 8$ spatial view transformations yields modest direction AUROCs of 0.539 for 3D UX-Net and 0.570 for SwinUNETR.
- **Conditional Density Overlap**: Figure 7 visualizes conditional entropy density distributions $p(H \mid P(v) = 1)$ and $p(H \mid N(v) = 1)$, showing near-identical overlap.

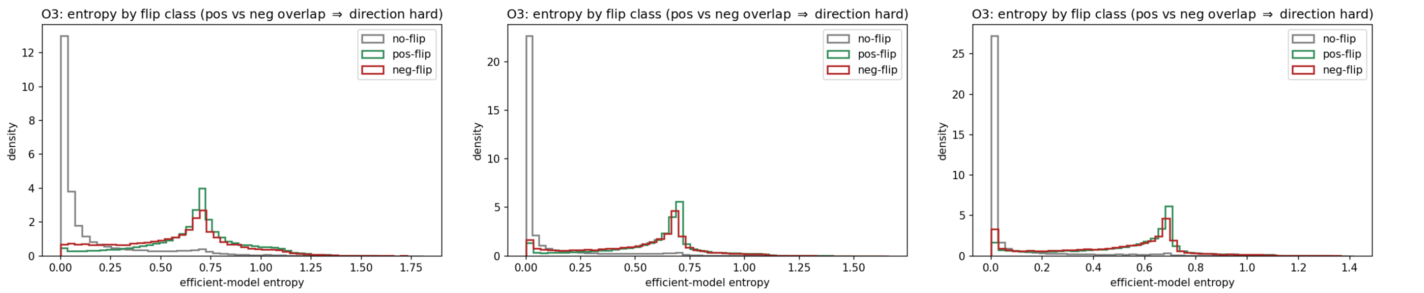
Table 4: 2×2 Factorial analysis of decoder channel reduction (C_{dec}) and resolution scaling (RF) on 3D UX-Net and BTCV CT (Decoder-Isolated Factor Attribution Axis).

Configuration	C_{dec}	RF	**Params (M)**	**FLOPs (GMac)**	**Latency (ms)**	**Mean Dice (%)**
Full (C_{384}, RF_1)	384	1	46.72	657.76	43.3	80.44
Chan-Only (C_{48}, RF_1)	48	1	2.24	388.15	35.2	79.80
Res-Only (C_{384}, RF_2)	384	2	46.32	319.10	16.1	77.97
Effi (C_{48}, RF_2)	48	2	1.84	49.49	8.0	78.35



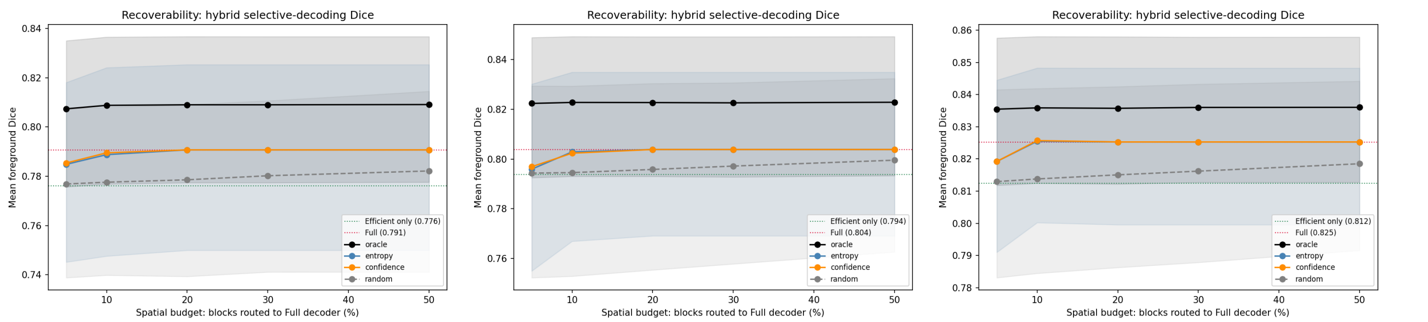
(a) 3D UX-Net Oracle Concentration (b) SwinUNETR Oracle Concentration (c) MedNeXt-M-K3 Oracle Concentration

Figure 6: Oracle positive-gain concentration curves across backbones. The dashed line represents the random selection baseline. In all three architectures, the positive gain is heavily skewed, satisfying $K_{80} \leq 5\%$.



(a) 3D UX-Net Flip Entropy Density (b) SwinUNETR Flip Entropy Density (c) MedNeXt-M-K3 Flip Entropy Density

Figure 7: Conditional entropy density distributions for positive corrections (P) and negative degradations (N).



(a) 3D UX-Net Macro Recovery (b) SwinUNETR Macro Recovery (c) MedNeXt-M-K3 Macro Recovery

Figure 8: Signal-guided retrospective macro-metric Dice recovery curves (R_{recovery}) under output hybridization vs. random selection.

5.5.1. Signal-Guided Retrospective Hybrid Recovery

Figure 8 presents retrospective hybrid prediction recovery curves $R_{\text{recovery}}(k)$ using various predictive signals:

- At a **5% block budget**, entropy-guided output substitution recovers **60% of the full decoder's Dice gap** over E_1 for 3D UX-Net (vs. 8% for random selection).

Table 5: Factor main effects (Δ_C, Δ_{RF}) and interaction ($\Delta_{C\times RF}$) across metrics on 3D UX-Net (BTCV CT).

Outcome Metric	Channel Δ_C	Stage Δ_{RF}	Interaction $\Delta_{C\times RF}$
Mean Dice (%)	−0.13	−1.96	+1.02
Params (M)	−44.48	−0.40	+0.00
FLOPs (GMac)	−269.61	−338.66	+0.00
Boundary Net Gain (%)	−0.020	+0.033	−0.004

Table 6: Inference-time predictability of block-level correctness activity (Location AUROC) and directional utility (Direction AUROC) across evaluated backbones.

Backbone	Location AUROC	Direction AUROC
3D UX-Net	0.900	0.591
SwinUNETR	0.913	0.560
MedNeXt-M-K3	0.907	0.556

- At a **20% block budget**, entropy-guided output substitution achieves **full recovery of the macro-Dice gap** for 3D UX-Net and SwinUNETR, while recovery remained partial for MedNeXt-M-K3 (vs. 17–20% for random selection).

5.6. Cross-Dataset Replication

To evaluate whether the observations discovered on BTCV generalize across imaging modalities and anatomical targets, Table 7 summarizes these metrics across all three datasets for the 3D UX-Net reference architecture. Across datasets, gross bidirectional correctness activity, positive-gain concentration (e.g., the strong concentration pattern persisted on MSD08 with $K_{80} \leq 5\%$), and the activity–direction predictability gap remained observable. However, the magnitude and sign of net gain were task-dependent: on MSD08, the efficient decoder outperformed the full decoder, producing a negative foreground-union net gain.

6. Discussion

This study shifts the evaluation of decoder efficiency from global FLOPs reduction to spatial computation dynamics. This section explains the underlying mechanisms, addresses the metric weighting paradox, outlines architectural implications, and discusses limitations.

6.1. Decoder Activity versus Net Benefit

Our findings show that additional decoder capacity does not simply act as a monotonic accuracy enhancer. Instead, it generates substantial gross correctness activity $A_{\text{corr}}(v) = P(v)+N(v)$, which is largely offset by bidirectional cancellation ($P(v) \approx N(v)$). Positive corrections and negative degradations coexist within ambiguous boundary regions, producing substantial local activity but much smaller aggregate net gain. Rather than systematically fixing errors volume-wide, high-capacity decoders reallocate predictions near ambiguous boundaries.

6.2. Coexistence of Small Voxel Net Gain and Macro Dice Improvement

A key question arises: *How can additional decoder computation yield a small global net voxel gain ($R_{net} < 0.1\%$) while still improving macroscopic Dice by 1–2 percentage points?*

This coexistence stems from the fundamental difference in metric weighting:

1. **Voxel Accuracy Weighting**: Global voxel correctness is heavily dominated by vast background regions and large organ interiors, where predictions are stable under both full and efficient decoders.
2. **Macro Dice Sensitivity**: The macro-averaged Mean Dice assigns equal weight to anatomical classes and is foreground-sensitive. Positive voxel corrections $P(v)$ occurring at thin anatomical boundaries or small, morphologically volatile organs (such as the adrenal glands) exert a disproportionately large impact on macroscopic Dice scores.

Therefore, a small global net voxel gain does not imply that additional decoder capacity lacks segmentation value; rather, its value is highly concentrated at Dice-sensitive boundary regions.

6.3. Implications for Adaptive Decoder Design

These findings suggest three practical directions for adaptive decoder design:

1. **Directly Predict Accuracy Gain**: Predictive entropy measures task difficulty and spatial ambiguity (> 0.90 Location AUROC), but provides weak directional discrimination (≈ 0.556–0.591 Direction AUROC). Future routing policies should train auxiliary heads to directly predict net accuracy gain rather than reusing task uncertainty signals.
2. **Decouple Spatial Resolution from Channel Capacity**: In the 3D UX-Net/BTCV factorial setting, retaining the high-resolution decoding stage primarily improved boundary-adjacent regions, whereas channel reduction produced a more spatially diffuse effect. This suggests exploring adaptive decoders that selectively allocate high-resolution refinement near anatomical boundaries while maintaining narrower channel widths elsewhere.
3. **Exploit Spatial Concentration for Compute Allocation**: Because positive gain is tightly concentrated in 5% of spatial blocks ($K_{80} \leq 5\%$), future dynamic compute allocation engines have a high theoretical performance ceiling if they can correctly identify these regions.

6.4. Limitations

This study has several limitations. First, the controlled interventions follow the channel-reduction and high-resolution stage-omission principles of EffiDec3D, and factor-level attribution is performed only on 3D UX-Net and BTCV. Although the end-to-end observations are evaluated across multiple architectures and datasets, the factorial effects may differ under other decoder designs or efficiency interventions.

Table 7: Cross-dataset generalization validation using 3D UX-Net across three clinical imaging datasets (Cross-Dataset Consistency Axis).

Dataset	**Full Dice** (%)	**Effi Dice** (%)	**Activity** $R^{\mathrm{all}}_{\mathrm{act}}$ (%)	**Net Gain** $R^{\mathrm{all}}_{\mathrm{net}}$ (%)	**Activity** $R^{\mathrm{fg}}_{\mathrm{act}}$ (%)	**Net Gain** $R^{\mathrm{fg}}_{\mathrm{net}}$ (%)	**Oracle** K_{80}	**Location AUROC**	**Direction AUROC**
BTCV CT	79.18	77.73	3.42	+0.046	12.84	+0.381	$\leq 5\%$	0.900	0.591
MSD01 BrainTumour	*[Pending]*	*[Pending]*	*[Pending]*	*[Pending]*	*[Pending]*	*[Pending]*	*[Pending]*	*[Pending]*	*[Pending]*
MSD08 HepaticVessel	57.51	58.63	0.20	−0.002	22.99	−4.89	$\leq 5\%$	0.865	0.520

Second, the end-to-end full and efficient models are trained independently under matched protocols. The frozen shared-encoder experiment controls variation in feature representations, but independent decoder initialization and optimization trajectories may still contribute to prediction differences.

Third, the routing analysis is based on retrospective output-space hybridization. It therefore provides an opportunity bound on selective decoder allocation rather than demonstrating a deployable dynamic execution system or proportional reductions in runtime latency and hardware-level computation.

Finally, predictability is evaluated primarily using predictive entropy and TTA mutual information on three public benchmarks. Broader clinical cohorts and learned or feature-based utility predictors may reveal additional information about directional decoder gain.

7. Conclusion

We systematically characterized decoder computation in 3D medical image segmentation. By evaluating matched full and efficient decoder pairs across representative 3D backbones and clinical datasets, we found that additional decoder capacity produces substantial gross correctness activity. However, this activity is largely offset by bidirectional cancellation, yielding much smaller and task-dependent net voxel shifts. Spatially, positive gain is highly concentrated near ground-truth anatomical boundaries ($K_{80} \leq 5\%$). At inference time across the BTCV backbones, predictive entropy reliably locates active regions (0.900–0.913 Location AUROC) but provides weak directional discrimination (0.556–0.591 Direction AUROC); these predictability gaps persisted on MSD08 despite a general reduction in AUROC magnitudes. Retrospective output-space hybridization on BTCV recovered the full macro-Dice gap for 3D UX-Net and SwinUNETR on a 20% block budget, while recovery remained partial for MedNeXt-M-K3. These results clarify how modern decoders behave spatially and provide empirical baselines for developing utility-aware adaptive decoders.

CRediT authorship contribution statement

Author One: Conceptualization, Methodology, Writing – original draft. **Author Two**: Data curation, Software. **Author Three**: Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study (BTCV/Synapse, MSD01 BrainTumour, and MSD08 HepaticVessel) are publicly available.

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Supplementary Material: Qualitative Visualizations for 3D UX-Net and MedNeXt-M-K3

This section provides supplementary qualitative visualizations for 3D UX-Net and MedNeXt-M-K3 backbones, complementing the SwinUNETR qualitative motivation example in Figure 2 of the main manuscript.

Supplementary Material: Entropy Monotonicity with Block-Level Error Rate

Figure S3 visualizes the block-level spatial correlation between mean predictive entropy and block-level error rate across all three evaluated backbones. The monotonic relationship confirms that entropy reliably localizes active, error-prone regions (Location AUROC), even though its directional discrimination ability remains weak.

Supplementary Material: Additional Implementation Details

TTA-MI (Test-Time Augmentation Mutual Information) Setup

For evaluating the predictive power of multi-view epistemic disagreement against spatial entropy, we utilized 8-view Test-Time Augmentation (TTA). Each volume was forwarded through the network across all 8 spatial flip combinations (x, y, z axes). The Mutual Information (TTA-MI) was computed as the difference between the entropy of the mean prediction and the mean of the individual view entropies. Despite this computationally expensive multi-pass estimation, TTA-MI provided only marginally better directional discrimination (Direction AUROC 0.570 for SwinUNETR, 0.539 for 3D UX-Net) compared to single-pass predictive entropy, further reinforcing our conclusion that uncertainty measures are powerful activity indicators but weak directional utility classifiers.

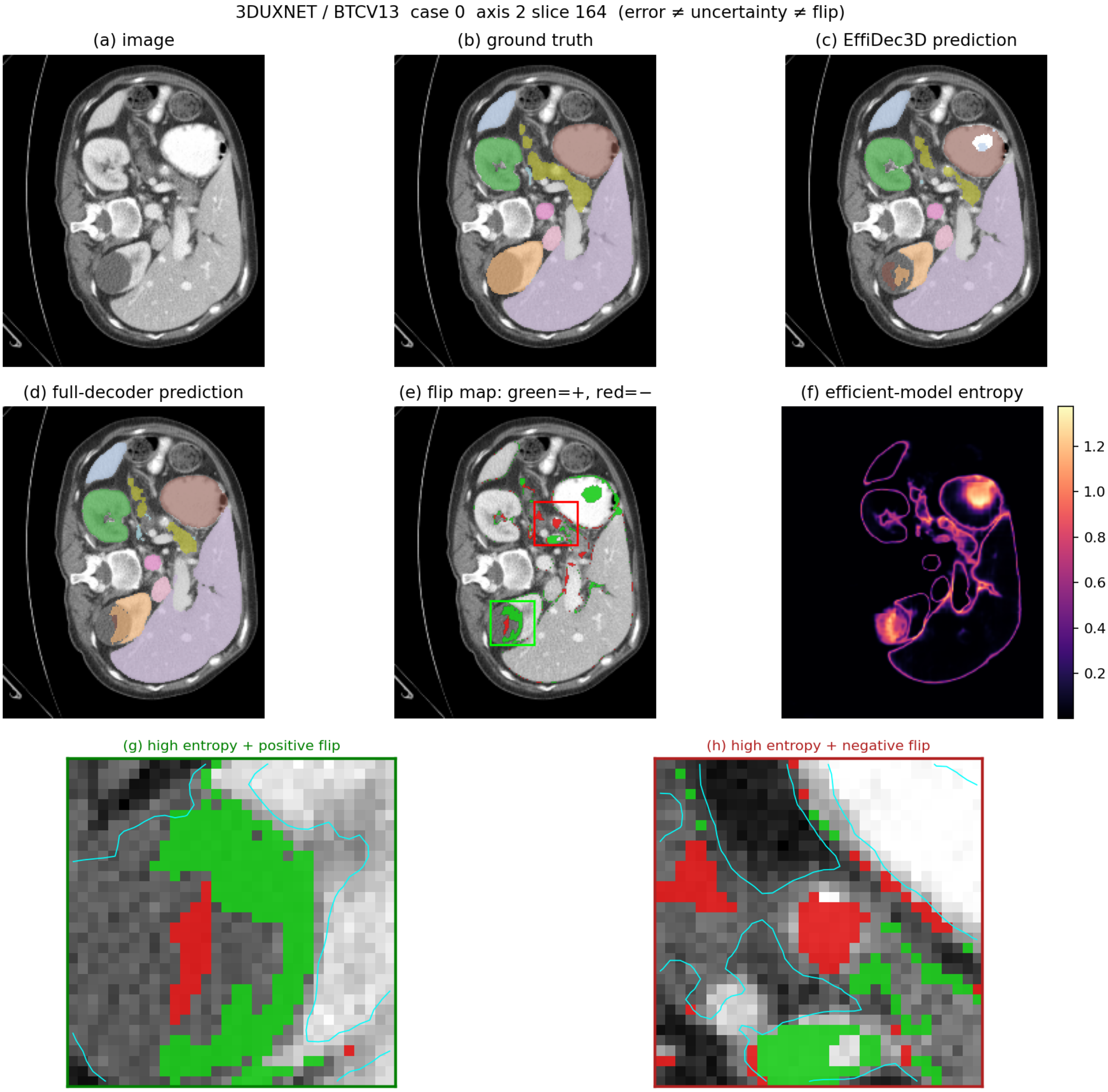


Figure S1: Supplementary Figure S1: Qualitative motivation visualization for 3D UX-Net on BTCV CT multi-organ dataset under Full (E_0) vs. Efficient (E_1) decoders.

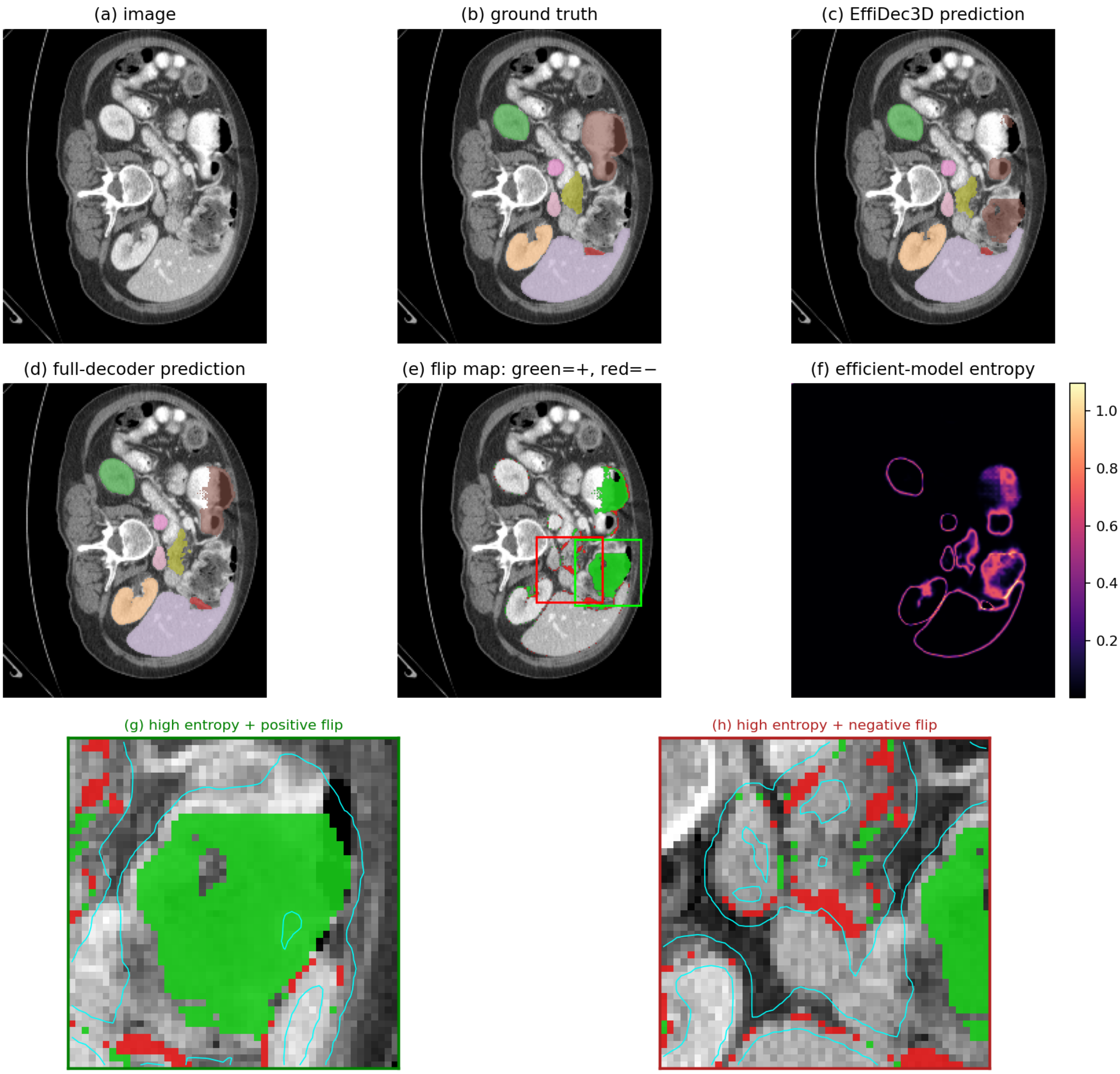
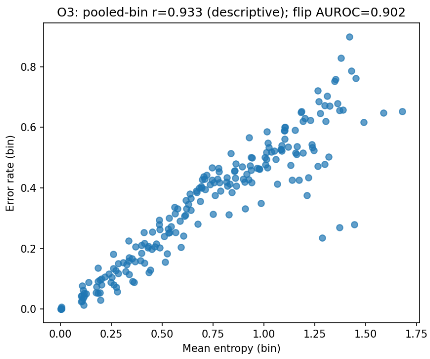
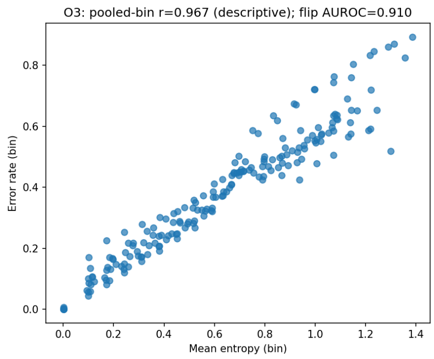


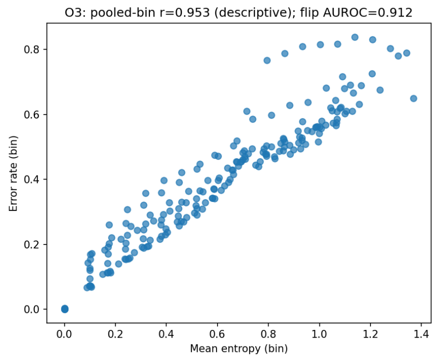
Figure S2: Supplementary Figure S2: Qualitative motivation visualization for MedNeXt-M-K3 on BTCV CT multi-organ dataset under Full (E_0) vs. Efficient (E_1) decoders.



(a) 3D UX-Net Error Correlation



(b) SwinUNETR Error Correlation



(c) MedNeXt-M-K3 Error Correlation

Figure S3: Supplementary Figure S3: Correlation between mean predictive entropy and block-level error rate across backbones.